Paleomagnetism of the Karmutsen basalts from southeast Vancouver Island

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One hundred and twenty oriented drill cores were collected from 20 sites in the Karmutsen basalts of Karnian age in the eastern part of Vancouver Island. Alternating field and thermal demagnetization studies reveal stable endpoint directions in 13 sites which fall into the following groups: (1) northwest-up (four sites); (2) north-up (four sites); (3) north-northeast-down (four sites); and (4) easterly-up (one site). The northwest-up direction is observed in the least altered samples with only pure magnetite of high coercive force and blocking temperatures as a carrier. The north-up direction, which shows streaking in a east–west sense, is also observed in (other) least altered specimens, and shows a complex magnetic mineralogy with lower coercivity. The north-northeast-down direction occurs in definitely more altered specimens, and some specimens show a northerly flat magnetization at the end of the blocking temperature range. The easterly-up direction is poorly defined but may correspond to the direction reported for the Nicolai Greenstones in Alaska. The relation between these groups of directions is not clear, and only in a few specimens an indication for the separation of remanence components has been observed. The available evidence indicates that a northerly component is probably the oldest and that the northwest-up magnetization may represent this component. The pole position calculated for this magnetization is 90°E, 21°N which is about 40°S of the Triassic pole position for cratonic North America. The discrepancy would be larger still if the northwest-up direction is a single component secondary direction. The discrepancy can, of course, be explained by a northward shift of the Karmutsen from a source area near 10°N with or without a sinistral rotation.

On a recueilli cent vingt carottes de forage orientées provenant de 20 sites dans les basaltes de Karmutsen datant du Karnien dans la partie est de l'île de Vancouver. Des études de désaimantation thermique et par champ alternatif mettent en évidence des directions finales stables pour 13 des sites et on les regroupe comme suit: (1) nord-ouest et vers le haut (quatre sites), (2) nord et vers le haut (quatre sites), (3) nord-nord-est, vers le bas (quatre sites), et (4) est, vers le haut (un site). On observe la direction nord-ouest vers le haut dans les échantillons les moins altérés avec seulement de la magnétite pure de force coercitive et de températures de blocage élevées comme porteur. On observe aussi la direction nord vers le haut avec des directions secondaires dans le sens est–ouest dans (d'autres) spécimens très peu altérés et possédant une minéralogie magnétique complexe avec des coercitivités plus faibles. La direction nord-nord-est vers le bas se retrouve dans des spécimens nettement plus altérés et quelques échantillons montrent une aimantation plane vers le nord à l'extrémité du domaine des températures de blocage. La direction est vers le haut est mal définie mais pourrait correspondre à la direction déjà rapportée pour les roches vertes de Nicolai en Alaska. La relation entre ces groupes de directions n'est pas claire et seulement dans quelques spécimens existe-t-il des indications permettant la séparation des composantes de rémanence. Les preuves disponibles indiquent que la composante vers le nord est probablement la plus ancienne et que l'aimantation nord-ouest vers le haut pourrait représenter cette composante. La position du pôle calculée pour cette aimantation est de 90°E, 21°N, ce qui est environ 40° au sud de la position du pôle pour le craton de l'Amérique du Nord au Trias. L'écart serait encore plus grand si la direction nord-ouest vers le haut était une direction secondaire à composante unique. On peut, bien sûr, expliquer la différence par un déplacement vers le nord de la région de Karmutsen à partir d'une région source située près de 10°N avec ou sans rotation sénestre.

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Introduction

Mesozoic and Tertiary intrusions and volcanics of the North American Cordillera have been the subject of several paleomagnetic studies. In particular, the Upper Triassic (Karnian) Karmutsen basaltic volcanics of Vancouver Island have been

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studied by Symons (1971) and a small collection has been investigated by Irving and Yole (1972). These studies did not include detailed thermal demagnetization but yielded complex results and two or more magnetization directions. These were considered to suggest translation or relatively minor regional tilting (Symons 1971), or a large northward shift of Vancouver Island (Irving and Yole 1972). Hillhouse (1977) studied the similar Nicolai Greenstone and postulated a 3000 km northwest shift coupled to a 90° sinistral rotation. The data from Symons (1971) and Irving and Yole (1972) can also be interpreted to indicate a northwest shift and a sinistral rotation where the shift would have taken place along a paleo-Andreas fault-like zone.

Because of the great importance of paleomagnetic data in the understanding of the dynamics that formed the Cordillera, another sampling of the Karmutsen volcanics was undertaken in areas of least deformation and near-horizontal bedding. Also, the Nicola Formation of mainland British Columbia, which is also of Upper Triassic age, was sampled to allow comparison of remanence directions on Vancouver Island and the mainland. The remanence of the Nicola samples was not suitable for paleomagnetic study and for that reason, only the results for the Karmutsen samples are reported here.

Geology and Sampling

Basaltic lavas of the Karmutsen Formation underlie more than one third of the total area of Vancouver Island (Fig. 1). They are generally composed of a lower unit of pillow lava, a middle unit of pillow breccia and aquagene tuff, and an upper unit of thickly layered, commonly amygdaloidal, basaltic lavas. In places, the lower pillow lavas overlie a sequence of pelitic sediments with Middle Triassic fossils that are intruded by thick diabase sills. The lavas are generally overlain by Late Triassic limestone and pelitic sediments of the Quatsino and Parson Bay formations and this whole sequence of Triassic rocks forms the Vancouver Group. The age of the Karmutsen volcanics is thus well defined as early Late Triassic (early Karnian), possibly extending back into late Middle Triassic (Ladinian). The estimated thickness of the formation varies from place to place, with a maximum of about 7500 m. The chemical composition of the Karmutsen is predominantly tholeiitic.

In several places the Karmutsen volcanics can be seen to overlie volcanic and sedimentary rocks of the middle to late Paleozoic Sicker Group, but it is not certain that these older rocks are present continuously in the subsurface. The Vancouver Group is overlain by the Early Jurassic Bonanza Group and all these rocks are intruded by Early (to Middle?) Jurassic intrusions. These and the older rocks, in turn, are overlain unconformably by Late Jurassic and Cretaceous clastic sediments.

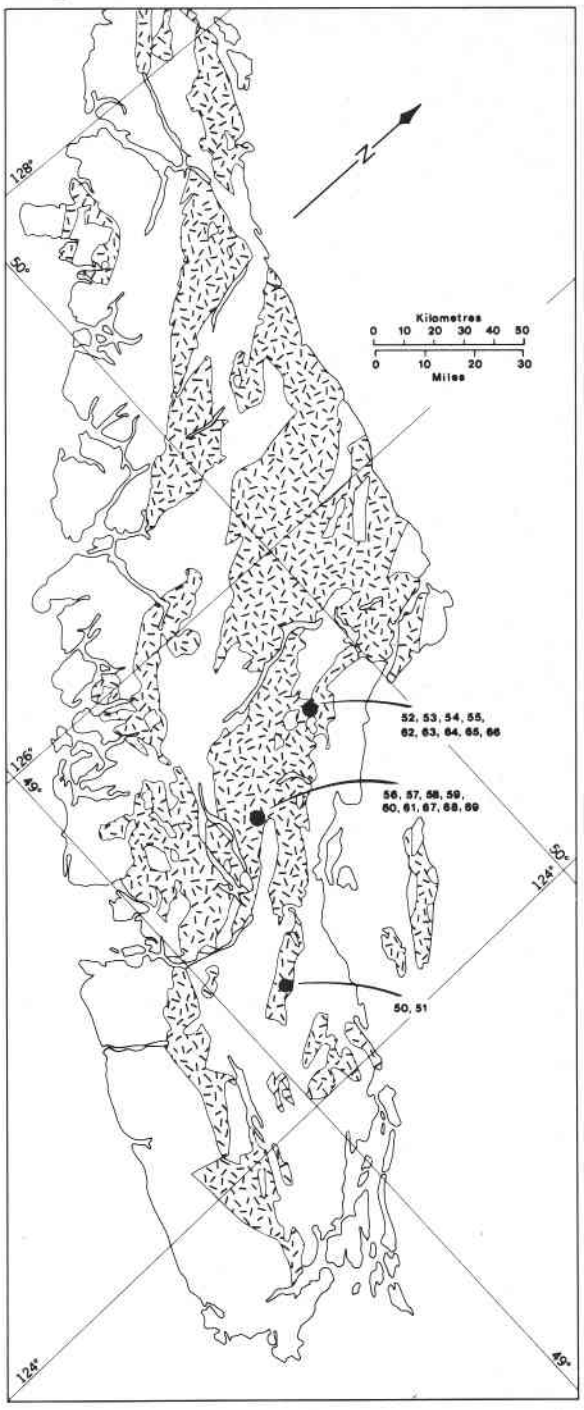


FIG. 1. Geological sketch map of Vancouver Island indicating extent of the Karmutsen Formation and the location of the sampling sites.

Muller (1977) has inferred that the Paleozoic volcanic arc terrane of the Insular Belt (including the Sicker Group of Vancouver Island) may be a displaced fragment of the Paleozoic geosyncline that was truncated and rifted off from the Pacific margin of southwest California and beyond in early Mesozoic time (as first noted by Hamilton and Myers (1966)). Consequently, the Karmutsen Formation may be credibly interpreted as the basaltic lavas extruded in the course of the rifting process.

Locations sampled for this study are in the upper part of the Karmutsen Formation, composed of layered flows, generally with slight apparent dip of the flow layers. Some locations are only a few kilometres distant from Jurassic Island Intrusions or Eocene Catface Intrusions (see Fig. 1). However, the grouping of remanent paleomagnetic directions established in this study does not seem to be related in any systematic way to their vicinity.

A total of 120 oriented drill cores were collected from 20 sites. Orientation usually was done with a sun compass; otherwise readings with a magnetic compass were checked by sighting prominent topographic features or other distant objects.

Remanent Magnetization

The oriented drill cores each yielded at least two cylindrical specimens with diameter and length of about 2.5 cm. Intensity and direction of the natural remanent magnetization (NRM) of each specimen were determined with an automated biastatic magnetometer or, when higher sensitivity was required, with a spinner magnetometer. The directions of the NRM were grouped per site as shown in Table 1. Many site directions are in the vicinity of the regional present geomagnetic field direction (declination, 23°E, inclination, +71°), which suggests the presence of a large recently acquired component. The NRM intensity range is 10^{-2}–10^{-5} emu (10^{-5}–10^{-8} SI). The grouping of the NRM directions of the majority of the sites is poor as shown by the low values of the precision parameter k.

To test the stability of the NRM, routine stepwise alternating field (AF) and thermal demagnetization were used. After each step, the direction and intensity of the remanence of each specimen were determined. The variation in intensity after AF treatment is shown on Fig. 2 for one specimen each of a number of sites and reflects a considerable range in the proportion of low and high coercivity remanence components. Samples with a large unstable component show a large rotation from downward to upward direction with a northerly azimuth (Fig. 2*b*). Other samples with downward magnetization show only a decrease in inclination, while still other samples do not show much change in direction, either upwards (e.g., 63) or downwards (e.g., 56), during AF treatment up to 1000 Oe (79 600 A/m).

Thermal treatment also shows large differences in results (Fig. 3 *a*, *b*). The thermal change in remanent intensity for specimens from the same site can be markedly different (e.g., 50 in Fig. 3*a*) in the proportion of low and high blocking temperature components. The highest blocking temperatures of most specimens were about 570°C, but some specimens show a minor component which is unblocked at 640–650°C. This corresponds with results obtained with a magnetic force balance which showed the presence of magnetite in all samples and very minor hematite in a few samples.

Directional changes are generally much more substantial during thermal demagnetization than during AF treatment where a large magnetization component survived fields up to 900 or 1000 Oe (71 640 or 79 600 A/m). Many specimen directions show swings either in a relatively low temperature range (one of the specimens from site 50 on Fig. 3*b*) or in a narrow temperature range (5–10°) near the high end of the blocking temperature spectrum (69, 60). These specimens contain at least one stable component which is directed northward. Upward directed components were also observed for a number of specimens such as one from sites 50 and 63 which do (63) or do not (50) show the presence of another major component. Some other specimens show much more complex behaviour such as a specimen from site 53 whose direction rotates in different temperature intervals but still ends up essentially horizontally north at the end of the blocking temperature spectrum.

Figure 4 shows all stable specimen directions (end points) obtained from the collection before (*a*) and after (*b*) rotating the flow planes at the various sites around strike to horizontal. Four groups of directions are distinguished on the basis of the data.

Northwest-up

These are specimens with either essentially single component magnetization, or, specimens which show evidence of single component magnetization after (partial) demagnetization in a wide range of higher AF field strength and temperature. This is shown by representative specimens in Figs. 2 and 3 and also by the essentially constant northwest-up direction of the vector removed during successive demagnetization steps (Fig. 5*a*). They usually show high coercivity components (Figs. 2*a* and 3*a*) and high stability is also borne out by the thermomagnetic work as discussed in a later paragraph. Four sites (50, 51, 62, 63), representing 23 cores or 38 specimens, show this direction as

TABLE 1. Site data before and after bedding plane rotation

Site	NRM				Treatment and results before rotation										After rotation	
	Total specimen No.	$D°$	$I°$	k	N	AF (Oe)	$D°$	$I°$	N	T (°C)	$D°$	$I°$	k	α_{95}	$D°$	$I°$
							Northwest-up									
50	12	341.5	52	4.38	8	300–500	323	−23	2	500–590			29.29	10.4	324.5	23.5
51	10	331.5	34.5	15.63	2	1000			6	570	320.5	−07	47.81	11.2	327.5	−21.5
62	12	327.5	02	4.13	8	300–500	323	−20	2				90.14	5.5	317.5	−17.5
63	12	313.5	−32	10.44	8	500–600	333	−24.5	2				33.19	9.8	341.5	−38
							North- to north-northeast-up									
58	12	014	05	10.46	8	500	004.5	−22.5	2				86.07	6.0	008	−41.5
59	12	002.5	20	18.33	8	500	002	−12	2	500			90.94	6.4	004	−31
64	8	357	−07.5	44.58	5	500	356.5	−22.5	1	640	347	−36	92.77	7.0	360	−25.5
65	12	347.5	18	7.02	2	1000			6	550	340	−26	42.65	10.4	347	−30
							North-northeast-down									
55	14	010	40.5	6.55	10	600–700	010	30.5	2	570			9.04	15.3	006	24
57	12	037.5	57	9.40	8	300	039	33	2				9.31	20.9	040	13
66	12	012	43.5	9.99	10	250	020.5	24	2	560	019	−04	7.20	22.2	008	23.5
69	12	013	55	26.57	2	650–700			8	560	017	48	55.99	7.5	028.5	51.5
							Easterly-up									
61	10	066.5	12	2.93	6	600	082.5	−40.5	2	640	101.5	−59	10.65	19.4	094	−35.5

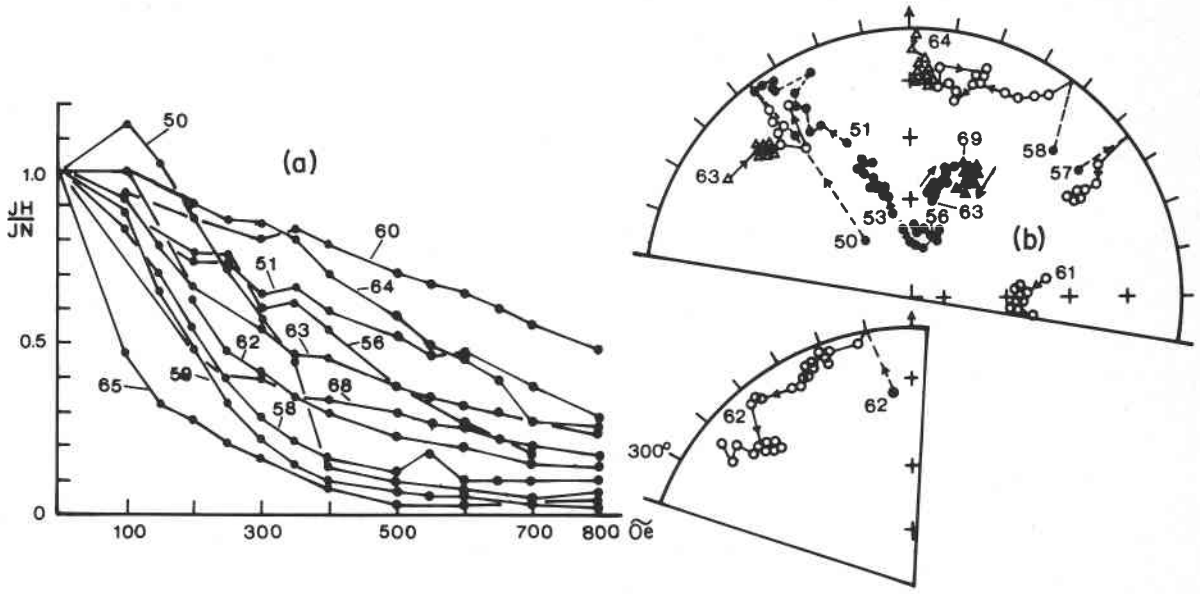


FIG. 2. Effect of alternating field treatment on NRM intensity (*a*) and direction (*b*) for a selection of samples. Numbers indicate site numbers. Open and closed symbols indicate directions in upper and lower hemisphere respectively. *JN* and *JH* indicate the intensity, respectively, before and after treatment.

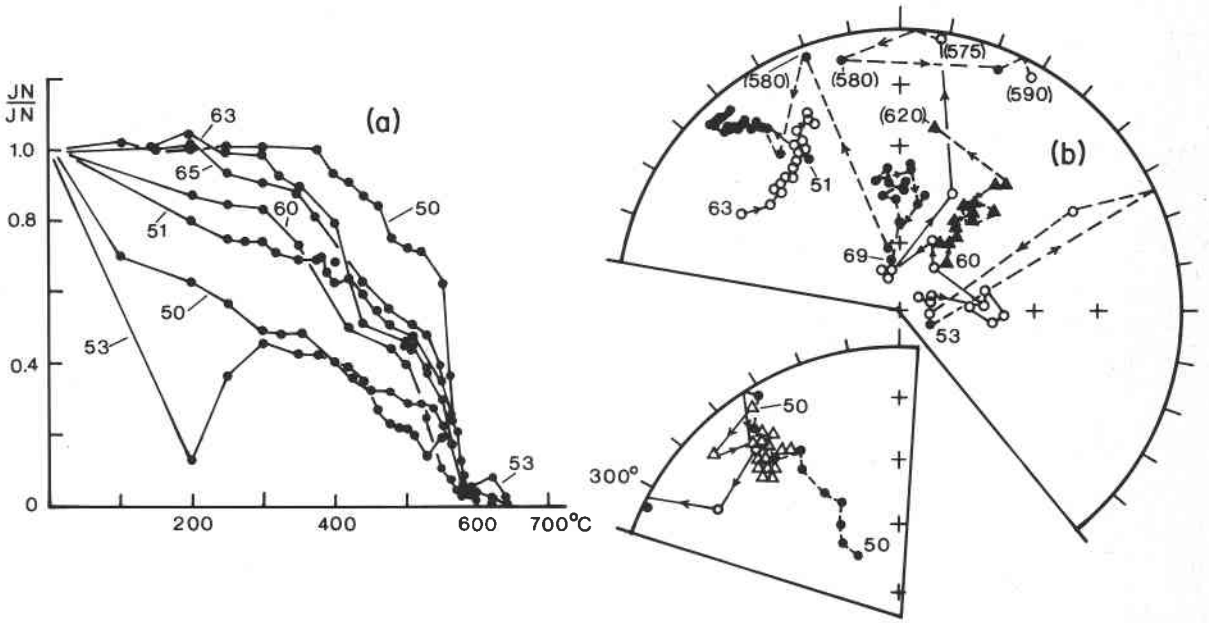


FIG. 3. Effect of thermal treatment on NRM intensity (*a*) and direction (*b*) for some samples. Bracketed numbers indicate temperatures. *JN* and *JT* indicate the intensity, respectively, before and after heating. Symbols as on Fig. 2.

endpoints (Table 1). Alternating field and (or) thermal treatment (see Table 1 for specifications) resulted in a strong increase in grouping of the endpoints. Rotating the gently dipping bedding planes to horizontal around their strikes does not appreciably change the grouping statistics as will be discussed in a later section. This direction has not been distinguished by either Symons (1971) or Irving and Yole (1972) in the Buttle Lake area.

North- to North-northeast-up

This direction forms part of the bimodal distribution of upward directions between northwest (see above) and north-northeast and is represented

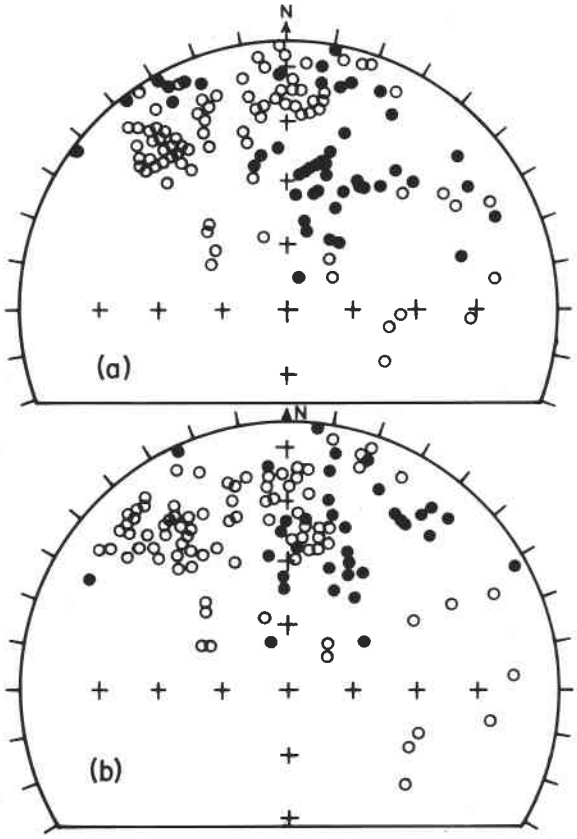


FIG. 4. Endpoints after AF and thermal treatment before (*a*) and after (*b*) rotation of flow planes to horizontal.

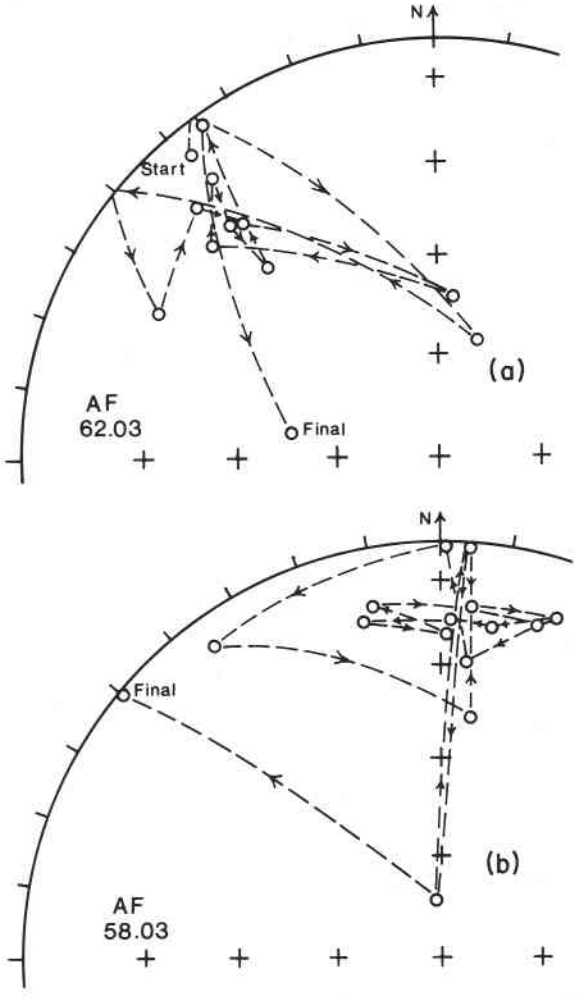


FIG. 5. Direction of the vector removed during successive steps of AF demagnetization for specimens of the northwest-up group of directions (*a*) and the north-up group (*b*).

by samples from sites 64, 58, and possibly 53 in Figs. 2 and 3. In this case too, essentially single component (64) as well as multicomponent magnetizations (58) occur. The group is clearly streaked towards the northwest-up direction and it may, therefore, not represent a single magnetization component. One sample (62 on Fig. 2) moves from north- to northwest-up, and in a few cases the vectors removed during successive treatments are aligned in this direction until the last stages of demagnetization when the vector removed direction has a tendency to move to northwest-up (Fig. 5*b*). Higher field thermomagnetic experiments also suggest that the stability is lower than that of the northwest-up magnetization. Four sites (58, 59, 64, 65), representing 22 cores or 34 specimens, show this north-up direction (Table 1). Again, AF and thermal treatment, as indicated on Table 1, showed a strong improvement in the direction grouping. In the present study the north-up direction is observed in samples from the Elsie Lake area but not farther to the east. This direction is reported in both earlier studies. On the basis of the pole position calculated from this direction, Irving and Yole (1972) have proposed a northward motion of Vancouver Island of more than 4000 km.

North-northeast-down

This magnetization is a high to medium coercivity magnetization as shown by the example in Fig. 2 and the higher field thermomagnetic experiments to be discussed in a later section. Thermal treatment is more effective and shows the existence of a northerly upward component of high blocking temperatures in several specimens but stable endpoints are not attained in these cases and motion does not closely define a great circle. This observation and the vicinity of the NRM directions to the present local geomagnetic field direction suggest that this remanence (represented by most samples from sites 55, 57, 67, and 69 and a few specimens from

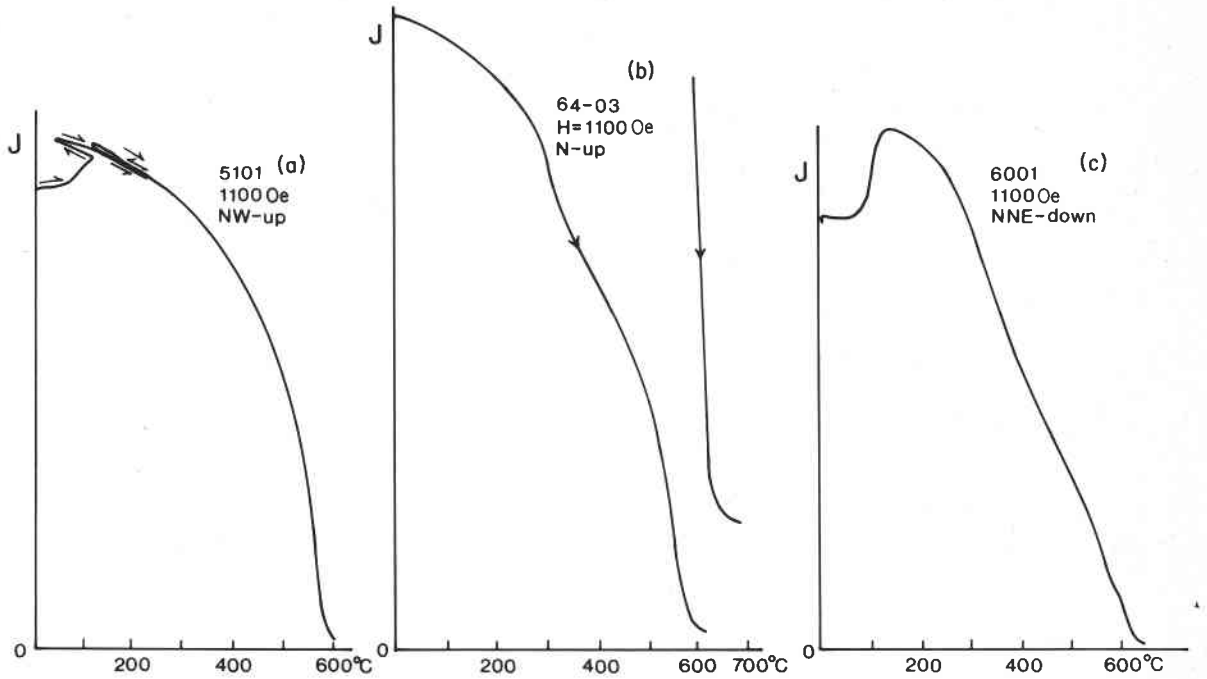


FIG. 6. Typical curves showing thermal change of the magnetization (J) obtained in moderate fields (H) and in saturating fields. The break of the curve in *b* is due to a 20 fold increase in sensitivity of the instrument.

other sites for a total of 36) might have been recently acquired by chemical alteration or crystallization of magnetic minerals. This direction has been reported by both Symons (1971) and Irving and Yole (1972).

Easterly-up

This direction is poorly defined by site 61 and a few individual specimens from other sites (57, 53). Coercivity is generally high but thermal treatment results, in a few cases, in qualitative evidence for the presence of a northerly direction (53 in Fig. 3*b*). This direction is similar to that reported by Hillhouse (1977) for the Alaskan Nicolai basalts, also of Triassic age, but these are thought to have been subjected to a northward movement of about 3000 km. This direction is obviously of subordinate importance in the present study, and it has been reported before by Symons (1971) to occur in one site.

Magnetic Mineralogy

Eight thin sections of sites from two sites each of the northwest-up, north-up, and north-northeast-down groups of magnetic directions were examined with a microscope. The rocks are plagiophyric basalts which sometimes contain augite phenocrysts. Large amygdules and interstitial openings are filled with chlorite, quartz, epidote, and prehnite, and suggest shallow water or subaerial extrusion. Metamorphism corresponds to pumpellyite–prehnite or low-greenschist facies. Pyroxene is generally replaced by amphibole. Magnetite is fairly abundant and is variably altered to sphene but in most cases alteration is minor.

In detail, the samples showing the north-northeast-down magnetization (e.g., site 69) display relatively strong alteration with all pyroxene replaced by amphibole. In contrast to this, all samples from sites 50 and 51 show only weak alteration of pyroxenes (northwest-up direction) with some oxydization of magnetite to possibly haematite. The samples from site 58 and 59 showing the north-up direction show very similar features to those from the northwest-up sites.

The magnetic mineralogy was also studied with a recording thermomagnetic balance. Specimens from the sites displaying northwest-up remanence directions showed only a single magnetic compound with a Curie temperature of 580°C which corresponds to that of pure magnetite (Fig. 6*a*). When the experiments were repeated in an applied field of 1100 Oe (87 560 A/m) instead of 5000 Oe (3.98×10^5 A/m), an initial increase in magnetization is observed which grows during thermal cycling (Fig. 6*a*). Since the Curie temperatures in the 5000 and 1100 Oe (3.98×10^5 and 87 560 A/m) field

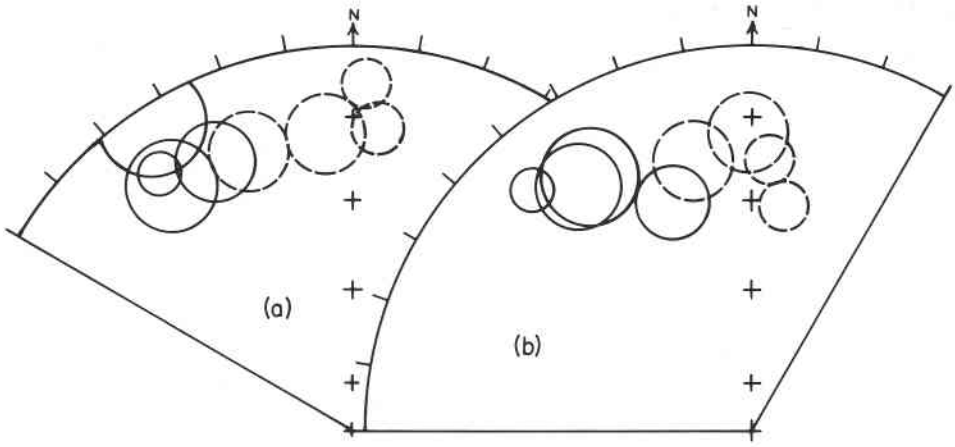


FIG. 7. Site means and their area of 95% confidence for four sites representing the northwest-up (solid lines) direction and four sites representing the north-up direction (dashed lines) before (*a*) and after rotation of flow planes to horizontal (*b*).

experiments are similar, this phenomenon is due to alignment of high coercivity remanence components in the applied field direction at slightly elevated temperatures which can be regarded as a viscous effect. Thus, high coercivity remanence of natural origin is present in these samples. Assuming rectangular hysteresis, the effective field H_{eff} equals the applied field minus the demagnetizing field: $H_{eff} = 1100 - NJ = 1100 - 3.5 \times 100 = 750$ Oe. This is a low estimate of the coercivity since the grain moment (J) is taken to equal the saturation moment at 20°C, while the demagnetizing factor (N) for an average magnetite grain is realistically set at 3.5. Thus, the maximum coercivity of the NRM is higher than 750 Oe (59 700 A/m).

The specimens from the north-up sites generally show a Curie temperature of about 580°C and a second one at about 350°C which may be due to a highly titaniferous magnetite or maghemite. This phase does not survive heating to 600°C and high coercivity features as described above have not been observed in any of these specimens (Fig. 6*b*).

Specimens from north-northeast-downward directed sites show up to three magnetic phases with Curie temperatures at about 350, 580, and 625°C. Only a few specimens show the high coercive force feature (Fig. 6*c*).

Interpretation

The first task in assessing the significance of these complex results is to determine at least the relative ages of the four groups of directions. The occurrence of the northwest-up magnetization in the least metamorphosed sites (prehnite–pumpellyite facies) and the high coercivity component detected in this magnetization can be used as a qualitative argument that this is an old, if not original, magnetization. Furthermore, the data indicate that this is a single component magnetization. The north-up magnetization also occurs in relatively fresh rocks but it does not contain a high coercivity component. Thus, it is possible that this is in fact a hybrid direction, one of the components of which is a relatively low-intensity northwest-up magnetization. The east–west streaking in this group of directions reinforces this interpretation. Furthermore, a few specimens from these sites move towards the northwest group of directions during progressive demagnetization, and a northwest component is removed only during the last stages of demagnetization but generally separation of components is not indicated by the data. The north-northeast-down magnetization can be regarded as a relatively young, sometimes high coercivity and high blocking temperature, magnetization as a result of directional swings towards a northerly direction displayed by some of the specimens in which the magnetization component was progressively destroyed during heating. This magnetization is consistently found in the most altered samples. The easterly-up magnetization is essentially exhibited by only one site and is, therefore, not discussed further.

The samples were collected from gently dipping flows in areas of relatively minor deformation. Dips in the Elsie Lake area were 15–20° while those farther east were 10° or less. Rotating the bedding planes around strike to horizontal, assuming their present attitude to be entirely secondary, or tectonic, yields the following observations (Fig. 7; Table 2):

(1) The average inclinations of the northwest-up and north-up groups of directions are increased significantly. The statistical parameters k and α_{95}, commonly used in paleomagnetism, do not show

consistent and significant changes if the population of individual sample directions is considered.

(2) The average inclination of the north-northwest-down group of directions is clearly decreased. The dispersion of the sample directions increases upon rotation but the concomitant increase in dispersion of the site directions is small. These observations indicate that rotation of the bedding planes does not yield conclusive results as to the age of the magnetization directions relative to any folding. A reason for this may well be that the present bedding attitude is partly primary.

The third factor which may have a bearing on the occurrence of different magnetization directions is the relative position of sampling sites in the stratigraphic column of the Karmutsen volcanics (Fig. 1). As seen from Fig. 1 and Table 1, the northwest-up direction is essentially confined to the easterly sampling areas in the relatively fresh upper part of the series, while the north-up direction occurs near Elsie Lake and Buttle Lake that may be stratigraphically lower. Thus, the north-up direction may be older than the northwest-up direction. However, as discussed earlier, there is evidence that the north-up direction may be a multicomponent magnetization which has been demonstrated in a few cases to contain a northwest-up component of high blocking temperature and coercivity. Furthermore, extrusion was probably relatively rapid about 210 Ma ago (Carlisle and Suzuki 1974), so that probably no more than a few million years elapsed between the beginning and end of this Karnian volcanic episode. Thus, directional differences due to apparent polar wander (~3°/Ma) and secular variation are expected to be small.

Pole Position

Figure 8 shows the paleomagnetic pole positions calculated for each of the four groups of directions both before and after rotating the bedding to horizontal. Added to this figure also is the average pole position, calculated by Hillhouse (1977), of pole positions determined for somewhat younger (190–200 Ma) rocks, mostly from eastern North America. The average pole position determined by Hillhouse (1977) for the Nicolai Group in Alaska is also shown.

Turning to the east-up direction first, its virtual pole falls very close to the Nicolai mean pole position. This would also be the case for Symons' (1971) site 22 from the lower Karmutsen in the Buttle Lake area. It seems, therefore, that this direction, which is so predominant in Hillhouse's data for the Alaskan Nicolai Greenstone, is sporadically observed in the Karmutsen of Vancouver Island. The significance of this observation

TABLE 2. Group mean statistics and pole positions

			Before flow plane rotation							After rotation							
							Pole position						Pole position				
Group	N	n	$D°$	$I°$	k	α_{95}	Long.	Lat.		$D°$	$I°$	k	α_{95}	Long.	Lat.	d_m	d_p
Northwest-up																	
Site	4		326.2	−18.5	82.0	10.2	91.9°E	24.0°N		327.2	−25.3	41.3	14.5	89.3°E	20.9°N	15.6	8.4
Sample		38	326.1	−19.5	30.9	4.7	91.8°E	23.5°N		326.9	−25.3	22.8	5.6	89.6°E	20.7°N	6.1	3.3
North-up																	
Site	4		355.7	−21.5	45.1	13.8	59.8°E	29.2°N		359.6	−32.3	65.1	11.5	55.4°E	22.9°N	12.9	7.3
Sample		34	356.5	−21.3	34.7	4.8	58.8°E	24.4°N		000.4	−33.0	41.2	4.4	55.6°E	22.5°N	5.0	2.8
North-northeast-down																	
Site	4		21.9	34.6	30.8	16.8	17.3°E	54.8°N		20.5	28.7	13.2	25.8	21.8°E	52.0°N	28.3	15.6
Sample		36	19.9	34.5	9.2	8.5	20.2°E	55.5°N		18.2	29.0	6.7	10.0	25.0°E	52.9°N	11.0	6.1
East-up																	
Site	1	8	92.0	49.7	10.6	19.4	165.5°E	12.8°N		93.8	−35.6	10.6	19.4	155.2°E	17.3°N	22.4	13.0

NOTES: Group statistics and pole position calculated on the basis of site directions (N is number of sites) and individual specimen directions (n is number of specimens). D and I indicate declination and inclination while k and α_{95} are the normally used precision parameters. d_m and d_p indicate the principal axes of the 95% oval of confidence around the pole position in degrees.

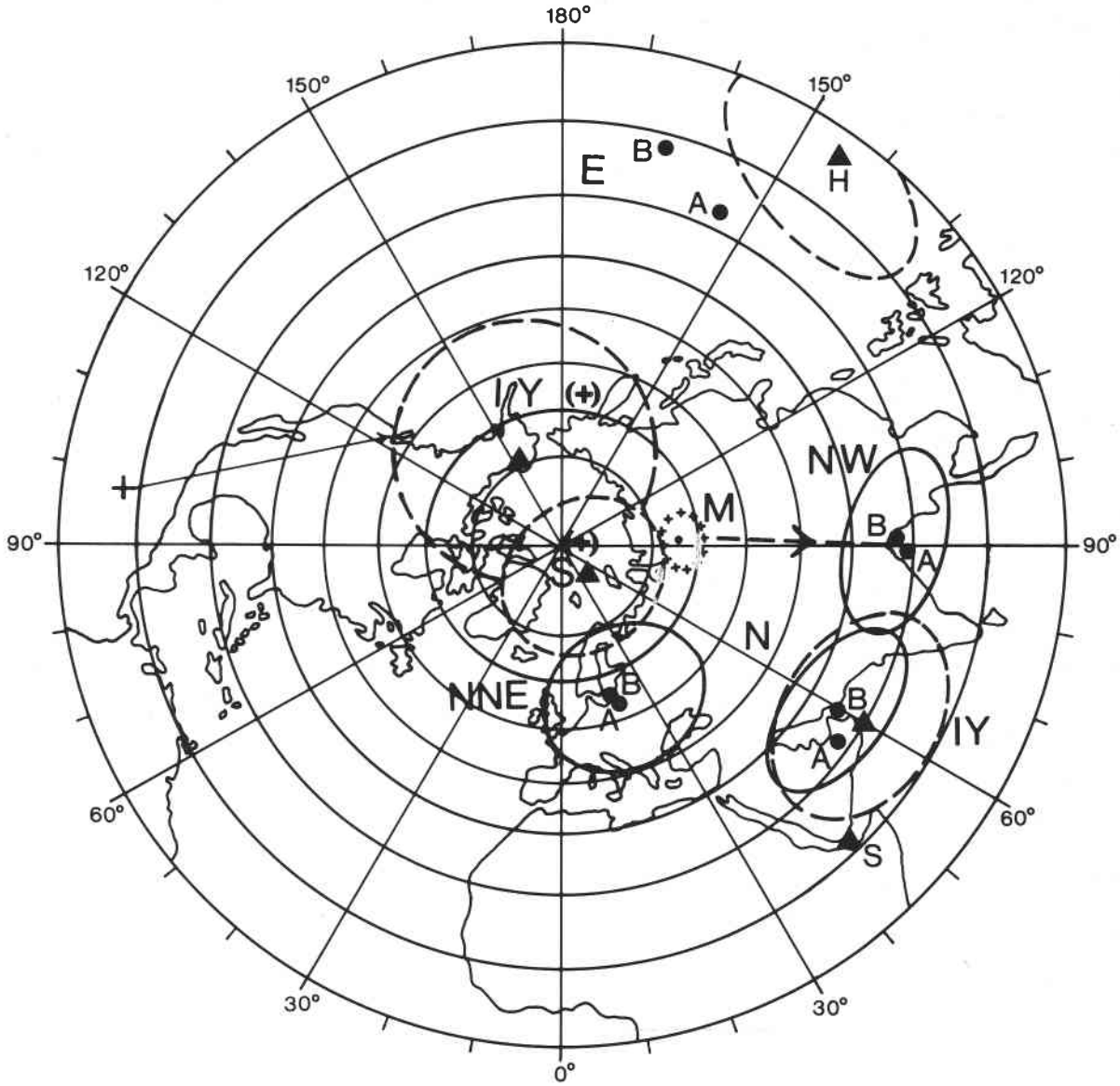


FIG. 8. Pole positions for the four magnetization directions distinguished before (B) and after (A) rotation of flow planes. Largest areas of 95% confidence from Table 2 are indicated. Nicolai Greenstone pole is labelled H; the results by Symons (1971) and Irving and Yole (1972) are indicated by black triangles identified by S and IY respectively, and the average North American cratonic pole position for Upper Triassic is indicated by M. Their area's of 95% confidence are represented by dashed lines. The cross and the arrow indicate a possible source area for Karmutsen and its net movement if poles A and B are to be brought into coincidence with the center of M in the most direct way.

is not clear but it may become of great interest if future work shows consistency in the occurrence of this direction.

The pole position derived from the north-northeast-down direction is approximately 20°S of the corresponding pole position reported by Symons (1971). The pole position derived from the north-up direction is about 20° northeast of the positions reported by Symons (1971) and similar to that of Irving and Yole (1972). These positions are to the southwest of the concentration of North American Upper Triassic and possibly Early Jurassic poles which has been interpreted as an indication for northward displacement coupled to a clockwise rotation. As discussed in an earlier part of this paper, considerable doubt must be placed on a single component nature of this direction and preference must, therefore, be given to the

northwest-up direction. However, this direction too yields a pole position which is south (by 43°) of the average position of North American cratonic poles as selected by Hillhouse (1977) on the other side of the globe. Thus, the northwest-up direction pole can be brought into coincidence with the average position by north–south displacement of Vancouver Island to a position of 123°W, 7°N without additional rotation. There are, of course, an infinite number of ways to get these poles to coincide by different combinations of northward shift and rotation which would result in other source areas for the Karmutsen volcanics but these must all include a large northerly directed component of motion.

The age of the northwest-up direction occurring in 20% of the sites has not been established, not even with respect to the folding. Only circumstantial evidence (high oxydation state, relatively simple magnetic mineralogy, high coercivity single component) suggests that it might be authentic. However, even if it is secondary, comparison with the post-Triassic part of the apparent polar wander curve for cratonic North America shows an even larger anomaly which then might suggest an even larger northward movement of the area in a shorter time span.

Conclusions

(1) The remanent magnetization of the Karmutsen basalts is even more complex than indicated by the earlier studies by Symons (1971) and Irving and Yole (1972). Four different groups of directions are indicated; (1) northwest-up, represented by 20% of the sites and observed exclusively in the eastern part of Vancouver Island which is probably underlain by the upper part of the series, (2) north-up, also represented by 20% of the sites, (3) north-northeast-down, shown by 20% of the sites, and (4) easterly-up, observed in 5% of the sites. The results for seven sites or 35% did not allow calculation of site direction, usually due to the absence of endpoints of the specimen directions. A few specimens falling in the north-northeast-down and north-up groups of directions show an indication for separation of components indicating a northerly- or northwesterly-up direction near the end of the blocking range.

(2) The age of the northwest-up direction could not be established but it is more likely than the other groups of directions to represent the original direction (least altered samples, pure magnetite as carrier with high coercivity and blocking temperatures).

(3) The pole position indicated by this direction falls far to the south of the Triassic and younger parts of the apparent polar wander curve for cratonic North America. This discrepancy may indicate a northward shift of Vancouver Island without a significant sinistral rotation or, for instance, by placing the source area of the basalts near the west coast of Mexico and allowing for such a rotation in addition to a northward movement.

(4) A poorly defined easterly-up direction is observed which may correspond to the direction generally observed in the Nicolai Formation of Alaska by Hillhouse (1977). This direction has formed the basis to postulate a large northwest movement coupled with a large sinistral rotation for Alaska. Further studies may indicate the distribution of this direction in areas between Alaska and Vancouver Island and show whether or not these easterly-up directions are contemporaneous. This would be important information because the Nicolai Greenstones are considered to be of similar age to the Karmutsen basalts.

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